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STM32 NUCLEO

Généralités

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Documentations techniques utiles

PM0056 Programming manual
STM32F10xxx/20xxx/21xxx/L1xxxxCortex-M3 programming manual

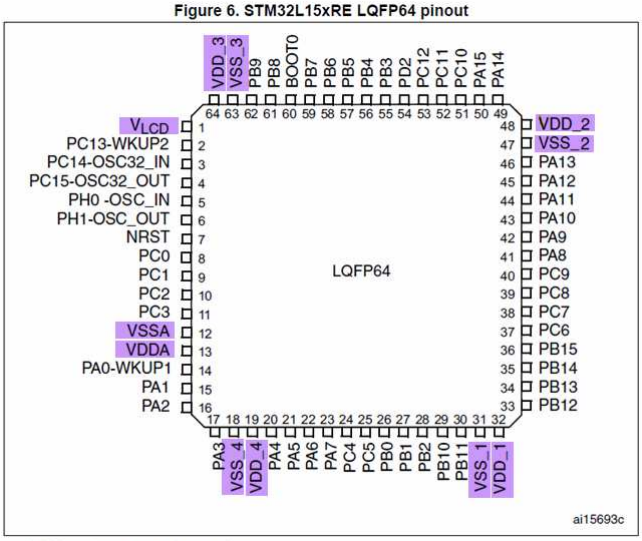
RM0038 Reference manual
STM32L100xx, STM32L151xx, STM32L152xx and STM32L162xx advanced ARM-based 32-bit MCUs

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Microcontrôleur STM32L152RE

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Figure 6. STM32L15xRE LQFP64 pinout



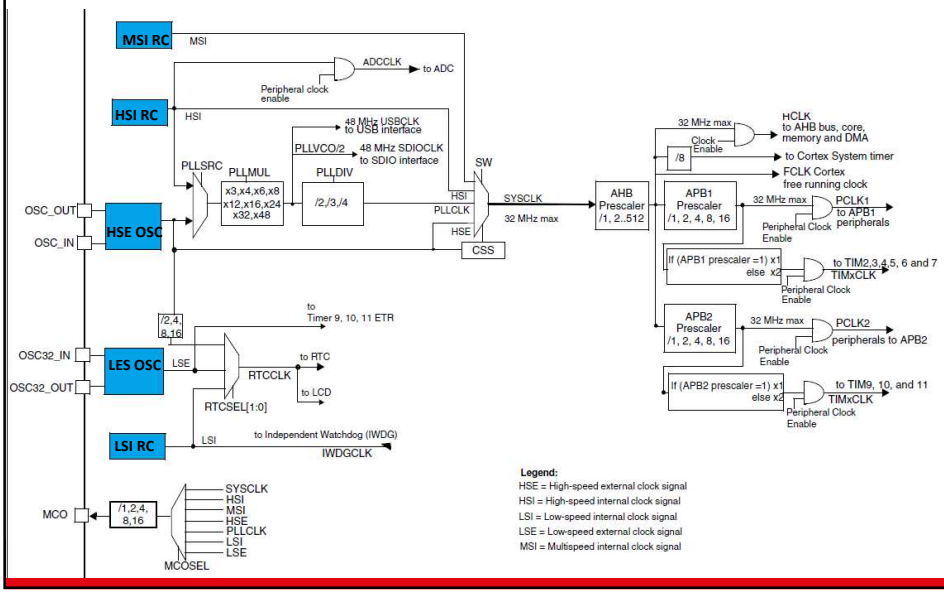
1. This figure shows the package top view.

Alimentation:
- VDD = 1,65 à 3,6V
- VSS = GND
- VDDA =VDD
- VSSA = VSS
- VLCD

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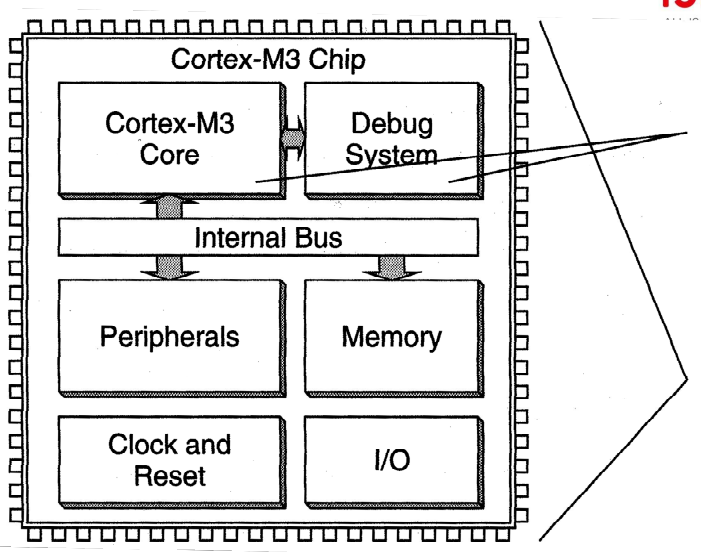
Arbre des Horloges du STM32L1X

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C quoi un µC ?

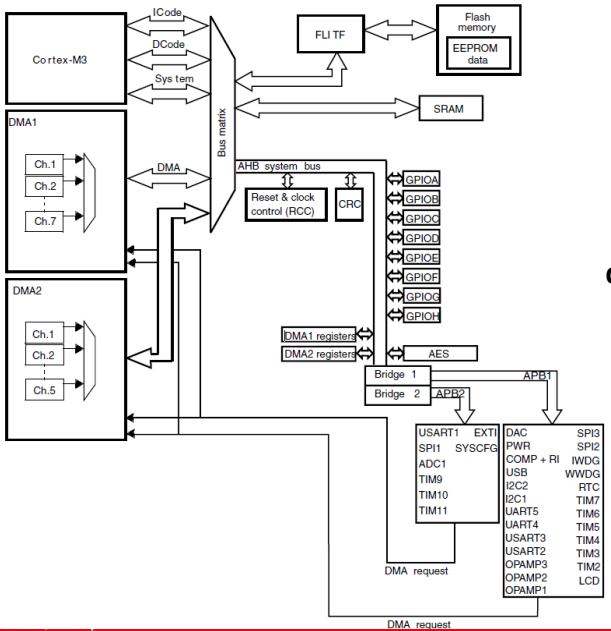
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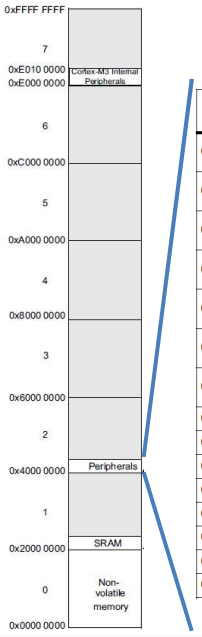
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Architecture du STM32L152RE

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Boundary address	Peripheral	Bus
0xA000 0000 - 0xA000 0FFF	FSMC	AHB
0x5006 0000 - 0x5006 03FF	AES	
0x4002 6400 - 0x4002 67FF	DMA2	
0x4002 6000 - 0x4002 63FF	DMA1	
0x4002 3C00 - 0x4002 3FFF	FLASH	
0x4002 3800 - 0x4002 3BFF	RCC	
0x4002 3000 - 0x4002 33FF	CRC	
0x4002 1C00 - 0x4002 1FFF	GPIOG	
0x4002 1800 - 0x4002 1BFF	GPIOF	
0x4002 1400 - 0x4002 17FF	GPIOH	
0x4002 1000 - 0x4002 13FF	GPIOE	
0x4002 0C00 - 0x4002 0FFF	GPIOD	
0x4002 0800 - 0x4002 0BFF	GPIOC	
0x4002 0400 - 0x4002 07FF	GPIOB	
0x4002 0000 - 0x4002 03FF	GPIOA	
……		

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Mapping de la mémoire

Activation de l'horloge sur les périphériques

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6.3.8 AHB peripheral clock enable register (RCC_AHBENR)

Address offset: 0x1C

Reset value: 0x0000 8000

Access: no wait state, word, half-word and byte access

Note: When the peripheral clock is not active, the peripheral register values may not be readable by software and the returned value is always 0x0.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	FSMC EN	Reserved		AES EN	Res.	DMA2EN	DMA1EN	Reserved							
	rw			rw		rw	rw								

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FLITF EN	Reserved		CRCEN	Reserved				GPIOG EN	GPIOF EN	GPIOH EN	GPIOE EN	GPIOD EN	GPIOC EN	GPIOB EN	GPIOA EN
rw			rw					rw	rw	rw	rw	rw	rw	rw	rw

Bit 25 **DMA2EN:** DMA2 clock enable

This bit is set and cleared by software.

0: DMA2 clock disabled

1: DMA2 clock enabled

Note: This bit is available in Cat.3, Cat.4 and Cat.5 devices only.

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6.3.9 APB2 peripheral clock enable register (RCC_APB2ENR)

Address: 0x20

Reset value: 0x0000 0000

Access: word, half-word and byte access

No wait states, except if the access occurs while an access to a peripheral in the APB2 domain is on going. In this case, wait states are inserted until the access to APB2 peripheral is finished.

Note: When the peripheral clock is not active, the peripheral register values may not be readable by software and the returned value is always 0x0.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	USART1 EN	Res.	SPI1 EN	SDIO EN	Res.	ADC1 EN	Reserved				TIM11 EN	TIM10 EN	TIM9 EN	Res.	SYSCF GEN
	rw		rw	rw		rw					rw	rw	rw		rw

Bits 31:15 Reserved, must be kept at reset value.

Bit 14 **USART1EN:** USART1 clock enable
This bit is set and cleared by software.
0: USART1 clock disabled
1: USART1 clock enabled

Activation de l'horloge sur les périphériques

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6.3.10 APB1 peripheral clock enable register (RCC_APB1ENR)

Address: 0x24

Reset value: 0x0000 0000

Access: word, half-word and byte access

No wait state, except if the access occurs while an access to a peripheral on APB1 domain is on going. In this case, wait states are inserted until this access to APB1 peripheral is finished.

Note: When the peripheral clock is not active, the peripheral register values may not be readable by software and the returned value is always 0x0.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
COMP EN	Res.	DAC EN	PWR EN	Reserved				USB EN	I2C2 EN	I2C1 EN	UART5 EN	UART4 EN	USART3 EN	USART2 EN	Res.
rw		rw	rw					rw	rw	rw	rw	rw	rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI3 EN	SPI2 EN	Reserved		WWD GEN	Res.	LCD EN	Reserved			TIM7 EN	TIM6 EN	TIM5 EN	TIM4 EN	TIM3 EN	TIM2 EN
rw	rw			rw		rw				rw	rw	rw	rw	rw	rw

Bit 31 **COMPEN:** COMP interface clock enable
This bit is set and cleared by software.
0: COMP interface clock disabled
1: COMP interface clock enable